

Foundations of Chain of thought reasoning

[Chain of thought reasoning]

1.0 Content Block

MIPRO (Multi-prompt Instruction Proposal Optimizer) optimizes the instructions and few-shot demonstrations of multi-stage language model programs, proposing small changes to module prompts and retaining those that improve a downstream performance metric without access to module-level labels or gradients. GEPA (Genetic-Pareto) is a reflective prompt optimizer for compound AI systems that combines language-model-based analysis of execution traces and textual feedback with a Pareto-based evolutionary search over a population of candidate systems; across four tasks, GEPA reports average gains of about 10% over reinforcement-learning-based Group Relative Policy Optimization (GRPO) and over 10% over the MIPROv2 prompt optimizer, while using up to 35 times fewer rollouts than GRPO. Open-source frameworks such as DSPy and Opik expose these and related optimizers, allowing prompt search to be expressed as part of a programmatic pipeline rather than through manual trial and error.

2.0 Content Block

Prompt engineering is the process of structuring natural language inputs (known as prompts) to produce specified outputs from a generative artificial intelligence (GenAI) model. Context engineering is the related area of software engineering that focuses on the management of non-prompt contexts supplied to the GenAI model, such as metadata, API tools, and tokens. It can also be defined as the practice of designing and refining input instructions given to a generative AI model to produce more accurate, relevant, or useful outputs. Effective prompt engineering involves understanding how a model interprets language, and may include techniques such as few-shot prompting, chain-of-thought prompting, and role assignment. It is increasingly considered a skill for working with large language models (LLMs) in both research and professional contexts. During the 2020s AI boom, prompt engineering became regarded as a business capability across corporations and industries. Employees with the title prompt engineer were hired to create prompts that would increase productivity and efficacy, although the individual title has since lost traction amid AI models that produce better prompts than humans and corporate training in prompting for general employees. Common prompting techniques include multi-shot, chain-of-thought, and tree-of-thought prompting, as well as the use of assigning roles to the model. Automated prompt generation methods, such as retrieval-augmented generation (RAG), provide for greater accuracy and a wider scope of functions for prompt engineers. Prompt injection is a type of cybersecurity attack that targets machine learning models through malicious prompts.

3.0 Content Block

In-context learning A model's ability to temporarily learn from prompts is known as in-context learning. In-context learning is an emergent ability of large language models. It is an emergent property of model scale, meaning that breaks in scaling laws occur, leading to its efficacy increasing at a different rate in larger models than in smaller models. Unlike training and fine-tuning, which produce lasting changes, in-context learning is temporary. Training models to perform in-context learning can be viewed as a form of meta-learning, or "learning to learn".

4.0 Content Block

Chain-of-thought (CoT) prompting is a technique that allows large language models (LLMs) to solve a problem as a series of intermediate steps before giving a final answer. In 2022, Google Brain reported that chain-of-thought prompting improves reasoning ability by inducing the model to answer a multi-step problem with steps of reasoning that mimic a train of thought. Chain-of-thought techniques were

developed to help LLMs handle multi-step reasoning tasks, such as arithmetic or commonsense reasoning questions. When applied to PaLM, a 540 billion parameter language model, according to Google, CoT prompting significantly aided the model, allowing it to perform comparably with task-specific fine-tuned models on several tasks, achieving state-of-the-art results at the time on the GSM8K mathematical reasoning benchmark. It is possible to fine-tune models on CoT reasoning datasets to enhance this capability further and stimulate better interpretability. As originally proposed by Google, each CoT prompt is accompanied by a set of input/output examples—called exemplars—to demonstrate the desired model output, making it a few-shot prompting technique. However, according to a later paper from researchers at Google and the University of Tokyo, simply appending the words "Let's think step-by-step" was also effective, which allowed for CoT to be employed as a zero-shot technique.

5.0 Content Block

Using gradient descent to search for prompts In "prefix-tuning", "prompt tuning", or "soft prompting", floating-point vectors are searched directly by gradient descent to maximize the log-likelihood on outputs. An earlier result uses the same idea of gradient descent search, but is designed for masked language models like BERT, and searches only over token sequences, rather than numerical vectors. Formally, it searches for $\arg \max_{\tilde{X}} \sum_i \log \Pr[Y_i | X \sim * \tilde{X}_i]$ where $X \sim$ ranges over token sequences of a specified length.

6.0 Content Block

Automated prompt generation Recent research has explored automated prompt engineering, using optimization algorithms to generate or refine prompts without human intervention. These automated approaches aim to identify effective prompt patterns by analyzing model gradients, reinforcement feedback, or evolutionary processes, reducing the need for manual experimentation.

7.0 Content Block

In 2022, text-to-image models like DALL-E 2, Stable Diffusion, and Midjourney were released to the public. These models take text prompts as input and use them to generate images. Early text-to-image models typically do not understand negation, grammar and sentence structure in the same way as large language models, and may thus require a different set of prompting techniques. The prompt "a party with no cake" may produce an image including a cake.

8.0 Content Block

Image prompting In 2023, Meta's AI research released Segment Anything, a computer vision model that can perform image segmentation by prompting. As an alternative to text prompts, Segment Anything can accept bounding boxes, segmentation masks, and foreground/background points.

9.0 Content Block

Tree-of-thought Tree-of-thought prompting generalizes chain-of-thought by generating multiple lines of reasoning in parallel, with the ability to backtrack or explore other paths. It can use tree search algorithms like breadth-first, depth-first, or beam.

10.0 Content Block

Context engineering Context engineering is a related process that focuses on the context elements that accompany user prompts, which include system instructions, retrieved knowledge, tool definitions, conversation summaries, and task metadata. Context engineering is performed to improve reliability, provenance and token efficiency in production LLM systems. The concept emphasises operational practices such as token budgeting, provenance tags, versioning of context artifacts, observability (logging which context was supplied), and context regression tests to ensure that changes to supplied context do

not silently alter system behaviour.